

Fig. 7 | Framework comparison

What the background-matched audit adds relative to prior MALDI-AMR publications and TRIPOD+AI.

Evaluation dimension	Prior MALDI-AMR	TRIPOD+AI	This framework
Reports external-site AUC	Yes	Required	Yes
Controls for co-resistance background	No	Not required	Yes
Quantifies signal collapse vs. retention	No	No	Yes
Site-specific evaluation	Yes	Recommended	Yes
Multi-drug panel (> 2 pairs)	Some	N/A	Yes
Second organism validation	No	N/A	Yes
Sensitivity analysis (stratum threshold)	No	No	Yes
Model-agnostic (accepts any CSV)	No	N/A	Yes
Open code + reproducible example data	Partial	N/A	Yes

Prior MALDI-AMR: Weis et al. 2022 (Nat. Med.) and comparable publications. TRIPOD+AI: transparent reporting guideline (2024).